THE UNIVERSITY OF HONG KONG
DEPARTMENT OF MATHEMATICS
MATH3603 Probability theory

Tutorial 5
2026/10/06

1. Let X and Y have the joint mass function

$$f(x,y) = \frac{C}{(x+y-1)(x+y)(x+y+1)}, \qquad x, y = 1, 2, 3, \ldots$$

Find the value of C, the mass function of $U = X + Y$, and the mass function of $V = X - Y$.

2. Let X and Y be independent random variables with common distribution function F and density function f. Find the distribution and density functions of $V = \max\{X, Y\}$, and those of $U = \min\{X, Y\}$.

3. Let X, Y and Z be independent and uniformly distributed on $[0, 1]$. Find the joint density function of XY and Z^2, and show that $\mathbb{P}(XY < Z^2) = \frac{5}{9}$.

4. Let X and Y be independent normal random variables, each having parameters μ and σ^2. Show that $X + Y$ is independent of $X - Y$. (**Hint**: Find their joint moment generating function.)